

Challenge (Habanero)**Open Ended Questions (FRQ)**

Q9. A park ranger is modeling the population of a certain animal species using the quadratic function $p(t) = at^2 + bt + c$. If the initial population is 50, and the population after 2 years is 70, find the values of a and b .

Q10. A hiker is traveling at a speed of $s(t) = 2t^2 + 5t + 1$ miles per hour. Find the time it takes for the hiker to travel 10 miles, and the maximum speed reached during the trip.

Chef's Tips

- Check for extraneous solutions
- Use the quadratic formula when necessary
- Show all work and explain reasoning